Face variation: Reputation-based Reward System & Manual Validation Framework

End-Phase 4 — Cycle 5 Updates

Phase 4, Cycle c5(End) – Oct 19th

Status of Manual Grading

We are officially done with manual review for Phase 4 — Cycle 5, and we are pleased with the results.

For this cycle, our goal was to **collect Real adversarial data for Passive Liveness**. With the data provided through the addresses shared for this cycle, we were able to release **Version 3 of Passive Liveness**

A major highlight from this cycle is the volume of data received. Active hotkeys grew from 159 to 204, and total submissions rose from 1,009,700 to 1,590,992, an increase of roughly 58%. Total challenges also grew, from 3,287 to 4,141. We also got more then **12,000** real screen replay image.

Scoring follows the same flow and criteria used to calculate the predicted scores shared in the last update. Overall, the final results were what was predicted. Where a score dropped from the predicted number, the similarity penalty was the reason. As with previous cycles, miners running more than one hotkey tended to see a larger similarity penalty than single-hotkey miners, which again shows that running multiple hotkeys is not in a miner's best interest.

We appreciate everyone who contributed data during Cycle 5. The data collected this cycle has directly contributed to **the improvement of Passive Liveness]**, and we look forward to building on it in the next cycle.

Metric	Phase 4 – Cycle 4	Phase 4 – Cycle 5
Hot Keys (Active)	159	204
Total Score	2,998.13	3,500.46
Total Challenges	3,287	4,141
Total Submissions	1,009,700	1,590,992
Positive Score Rate	73.65% (780,782 images)	69.59% (1,978,750 images)

Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 5 is:

Participation History	Age Factor	Bonus
Phase 3 — Cycle 1 (and active now)	1.50	+50%
Phase 4 — Cycle 1 (and active now)	1.40	+40%
Phase 4 — Cycle 2 (and active now)	1.30	+30%
Phase 4 — Cycle 3 (and active now)	1.20	+20%
Phase 4 — Cycle 4 (and active now)	1.10	+10%
Phase 4 — Cycle 5 (new miner)	1.00	+0%

Final Score Calculation

We're using the same formula as last cycle:

Total Score = ((Face Variation Score * 0.5) + (Screen replay score * 0.5))× Age Factor

Where:

- Face Variation Score is computed from the Δrep mapping table (unchanged, see below).

- The real Screen replay's sent by the miners
- Age Factor is based on the participation history described above.

New Tier (Extended Top-End Normalization)

Note any tier changes this cycle. For reference, the table is included again below:

Tier	Range of Score	Multiplier
🏅 Platinum	150–249.99	1.25
💎 Diamond	50–149.99	1.15
⭐ Gold	10–49.99	1.10
🥈 Silver	2.00–9.999	1.05
🥉 Bronze	1.00–1.999	1.02
☐ Neutral	0.70–1.00	1.00
‼ Watch	0.10–0.699	0.90

Decay

Decay will increase from 0.02-> 0.05

Leaderboard Growth

Leaderboard growth summary — compare Cycle 5 submissions and positive score rate against Cycle 4.

Metric	Phase 4 – Cycle 4	Phase 4 – Cycle 5
Total Submissions	1,009,700	1,590,992
Positive Score Rate	73.65%	69.59%

Now that final scoring (including the similarity penalty) is complete, the reputation leaderboard for Phase 4 — Cycle 5 holds. The tier breakdown for this cycle:

Tier	Cycle 4	Cycle 5

Platinum	1	1
Diamond	29	24
Gold	17	26
Silver	26	23
Bronze	37	18

Miner's Reputation Leaderboard (Top 10)

Reputation is determined by validated Face performance. Miners not listed here fall into the Neutral tail or the Watchlist, or did not score high enough to surface.

Rank	Hotkey	Score	Tier
1	5Hmz	180.12	Platinum
2	5C5h	129.08	Diamond
3	5GCX	126.83	Diamond
4	5FCD	109.10	Diamond
5	5EBx	104.00	Diamond
6	5CCe	102.66	Diamond
7	5DJk	86.12	Diamond
8	5DhK	72.39	Diamond
9	5DWm	71.43	Diamond
10	5CorH	68.37	Diamond

New Δrep Mapping Table

Validation Score	Meaning	Δrep
+6	Breaks system	+0.15
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03

+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
–1	Partial spam	-0.05
–2	Misleading	-0.08
–3	Duplicate	–0.10
–4	Not enough info	-0.25
–5	Cheat	-0.5

Top 2 Miners' Score Breakdown

The tables below show how raw validation outcomes translated into reputation for the top three miners this cycle.

5Hmz: Hotkey (Rank 1)

Face

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	0	0	7	11	146	148	703	715	1134	556	564	0	3984
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	0	0	-0.7	-0.88	-7.3	0	7.03	14.3	34.02	22.24	56.4	0	125.11

Screen replay

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	0	0	0	0	0	0	72	0	831	821	216	233	**2173**
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	0	0	0	0	0	0	0.72	0	24.93	32.84	21.60	34.95	**115.04**

Face score = 125.11

Screen replay score = 115.04

Last cycle score = 0

Age = 6 cycles

Face score + Screen score = (125.11 * 0.5) + (115.04 * 0.5) = 120.08

Age score = 120.08 × 1.6 = 180.12

Rep score = 180.12

5C5h: Hotkey (Rank 2)

Face

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	0	0	10	46	186	379	773	889	933	507	207	0	3930
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	0	0	-1.0	-3.68	-9.3	0	7.73	17.78	27.99	20.28	20.7	0	80.5

Screen

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	**0**	**0**	**0**	**0**	**0**	**0**	**13**	**3**	**902**	**567**	**338**	**229**	**2052**
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	**0**	**0**	**0**	**0**	**0**	**0**	**0.13**	**0.06**	**27.06**	**22.68**	**33.80**	**34.35**	**118.08**

Face score = 80.5

Screen replay score = 118.04

Last cycle score = 0

Age = 4 cycles

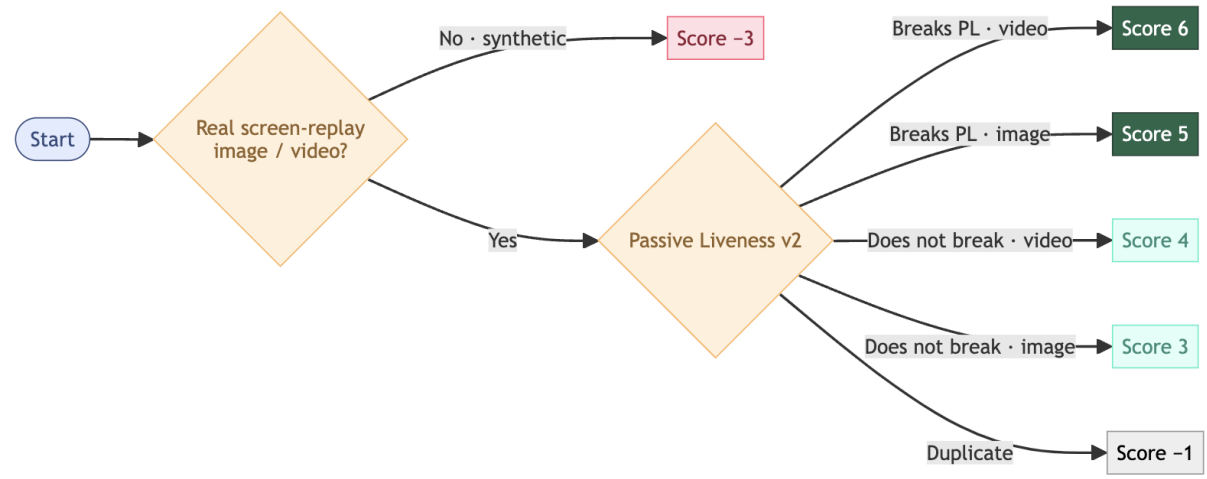
Face score + Screen score = (80.5 * 0.5) + (118.04 * 0.5) = 99.29

Age score = 99.29 × 1.3 = 129.08

Rep score = 129.08

How manual review scoring works

Flow



Score sheet

score	Reason
6	Video that broke our PL
5	Image that broke our PL
4	Video that did NOT break our PL
3	Image that did not break our PL

Issue Scoring Rules

Manual reviewer has a score sheet:

Rule	Score Impact	How It Applies
Single image only	-1	Apply when the required second image is missing, because the submission cannot be checked from two views.
Missing image description	-1	Apply when the required second image is missing, because the submission cannot be checked from two views.
Duplicate submission	Automatic −1	A duplicate receives an automatic score of −1. Exception: an accidental duplicate from the same hotkey receives 0 for one copy, while the other copy is reviewed normally.
Synthetic image	Automatic −3	A synthetic image receives an automatic score of −3 because the task explicitly requires real images.

End-Phase 4 — Cycle 4 Updates (Aug 23, 2026)

Phase 4, Cycle c4(End) – Aug 23, 2026

Status of Manual Grading

We are officially done with manual review for Phase 4 — Cycle 4, and we are pleased with the results.

For this cycle, our goal was to help miners provide adversarial data for our Passive Liveness system and BitMind AI Detection. With the data provided through the addresses shared for this cycle, we were able to make significant progress on Passive Liveness and develop Version 2 of the system.

A major highlight from this cycle is the volume and quality of the adversarial data received. We received more than 12,000 images that were able to break Passive Liveness, giving us valuable data to improve the system and make it more robust against these types of attacks.

Scoring follows the same flow and criteria used to calculate the predicted scores shared in the last update. Overall, the final results were very close to what was predicted. Where a score dropped from the predicted number, the similarity penalty was the reason. As with previous cycles, miners running more than one hotkey tended to see a larger similarity penalty than single-hotkey miners—another data point confirming that running multiple hotkeys is not in a miner's best interest.

We appreciate everyone who contributed data during Cycle 4. The adversarial examples collected this cycle have directly contributed to improving Passive Liveness and moving us toward a stronger Version 2.

Metric	Phase 4 – Cycle 3	Phase 4 – Cycle 4
Hot Keys (Active)	74	159
Total Score	2,596.89	2,998.13
Total Challenges	3,455	3,287
Total Submissions	413,657	1,009,700
Positive Score Rate	76.09% (314,771 images)	73.65% (780,782 images)

Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 3 is:

Participation History	Age Factor	Bonus
Phase 3 — Cycle 1 (and active now)	1.40	+40%
Phase 4 — Cycle 1 (and active now)	1.30	+30%
Phase 4 — Cycle 2 (and active now)	1.20	+20%
Phase 4 — Cycle 3 (and active now)	1.10	+10%
Phase 4 — Cycle 4 (new miner)	1.00	+0%

Final Score Calculation

We're using the same formula as last cycle:

Total Score = (Remaining Rewards from Phase 4 – Cycle 3) + (Face Variation Score × Age Factor)

Where:

- Remaining Rewards from Previous Cycle carries forward any unrealized rep_score from Phase 4 — Cycle 3.
- Face Variation Score is computed from the Δrep mapping table (unchanged, see below).

- Age Factor is `1.20`, `1.10`, or `1.00` based on the participation history described above.

New Tier (Extended Top-End Normalization)

This cycle introduces a new Black tier (rep score > 250, ×1.35) and shifts the top cutoffs so Platinum is 150–250 and Diamond is 50–150; Gold and below are unchanged. For reference, the table is included again below:

Tier	Range of Score	Multiplier
♣ Black	250+	1.35
🥇 Platinum	150–249.99	1.25
💎 Diamond	50–149.99	1.15
★ Gold	10–49.99	1.10
🥈 Silver	2.00–9.999	1.05
🥉 Bronze	1.00–1.999	1.02
☐ Neutral	0.70–1.00	1.00
‼ Watch	0.10–0.699	0.90

Decay

For the cycle, there will be a time decay where all hotkeys are going to decay at the same rate at the same time no matter if they're registered or not.

Decay will decrease from 0.05 -> 0.02

Leaderboard Growth

The previous cycle saw a significant increase in both output quality and volume, with submissions nearly quadrupling while the positive score rate improved substantially. Cycle 4 continued that momentum, with total submissions rising from **413,657 to 1,009,700**, a **144% increase**, while the positive score rate remained relatively strong at **73.65%**. Although the rate declined slightly from 76.09%, it remained broadly in the same range despite more than doubling the submission volume. This suggests that the substantial growth in activity did not come at the expense of overall quality:

Metric	Phase 4 – Cycle 3	Phase 4 – Cycle 4
Total Submissions	413,657	1,009,700
Positive Score Rate	76.09%	73.65%

Now that final scoring (including the similarity penalty) is complete, the reputation leaderboard for Phase 4 — Cycle 4 holds 159 miners. This shows an overall average increase in miner quality. More miners are getting higher tiers. The tier breakdown for this cycle:

Tier	Cycle 3	Cycle 4
Black	---	0
Platinum	1	1
Diamond	20	29
Gold	10	17
Silver	13	26
Bronze	11	37

Miner's Reputation Leaderboard (Top 10)

Reputation is determined by validated Face performance. Miners not listed here fall into the Neutral tail or the Watchlist, or did not score high enough to surface.

Rank	Hotkey	Score	Tier
1	5Hmz...BE63	165.42	Platinum
2	5FWd...1xfq	102.53	Diamond
3	5DWm...avYt	100.77	Diamond
4	5FCD...Z6fa	99.72	Diamond

5	5Cor...VtWA	99.08	Diamond
6	5C5h...kAQF	95.03	Diamond
7	5G1L...NBbD	94.67	Diamond
8	5C7X...aSVT	93.6	Diamond
9	5DiM...N1zM	87.28	Diamond
10	5Df9...eF5f	84.68	Diamond

New Δrep Mapping Table

Validation Score	Meaning	Δrep
+6	Breaks system	+0.15
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03
+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
–1	Partial spam	-0.05
–2	Misleading	-0.08
–3	Duplicate	–0.10
–4	Not enough info	-0.25
–5	Cheat	-0.5

Top 3 Miners' Score Breakdown

The tables below show how raw validation outcomes translated into reputation for the top three miners this cycle.

5Hmz...BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	0	0	77	43	53	112	83	298	892	1120	536	0	3214

Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	0	0	-7.7	-3.44	-2.65	0	0.83	5.96	26.76	44.8	53.6	0	118.16

Face score = 118.16

Last cycle score = 0

Age = 5 = 1.4 (age factor)

Age score = 118.16 × 1.4 = 165.42

Rep score = last cycle score + age score = 165.42

5FWd...1xfq

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	total
Count	0	1	14	2	11	26	23	62	101	44	5	0	289
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	0.15	
Total	0	-0.25	-1.4	-0.16	-0.55	0	0.23	1.24	3.03	1.76	0.5	0	4.4

Face score = 4.4

Last cycle score = 97.69

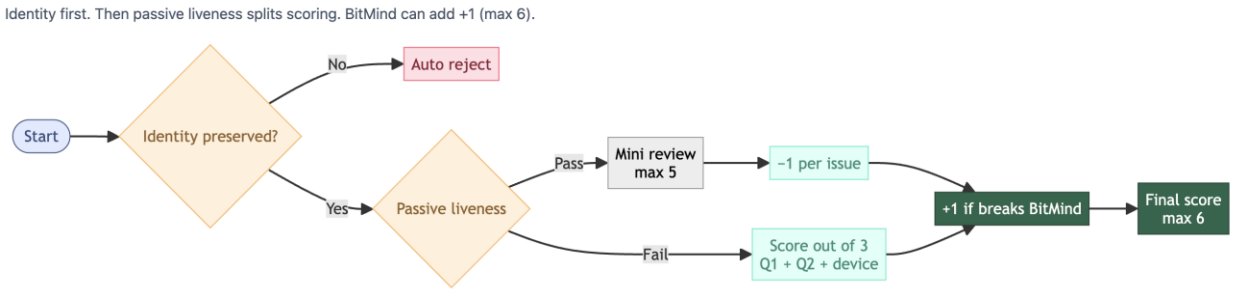
Age = 2 = 1.1 (age factor)

Age score = 4.4 × 1.3 = 4.84

Rep score = last cycle score + age score = 102.53

How manual review scoring works

Flow



Note: if image did not pass passive life max score is out of three instead of 5

Score sheet

Manual reviewer has a score sheet:

Score	# of issues
5	Clean — no issues
4	1 issue
3	2 issue
2	3 issue

Example

Issue type	Example	Comment
Not correctly AI-generated		Double bottom edge of screen

Face too small / not dominant		---
noise / screen board not there		Floating face, does not accurately reflect a screen replay
Low identity score	---	---

End-Phase 4 — Cycle 3 Updates (June 29, 2026)

Phase 4, Cycle c3(End) – Jun 29, 2026

Status of Manual Grading

We are officially done with manual review for Phase 4 — Cycle 3. We are pleased with the results and one of our goals is cycle was to help the miners and we saw an overall improvement with the miner scores.

Scoring follows the same flow and the same criteria used to calculate the predicted scores shared in the last update. Overall, final results are very close to what was predicted. Where a score dropped from the predicted number, the similarity penalty is the reason. As with previous cycles, miners running more than one hotkey tended to see a larger similarity penalty than single-hotkey miners — another data point confirming that running multiple hotkeys is not in a miner's best interest.

Metric	Phase 4 – Cycle 2	Phase 4 – Cycle 3
Cold Keys (Unique Miners)	26	24
Hot Keys (Active)	101	74
Total Score	889.27	2,596.89
Total Challenges	4,930	3,455
Total Submissions	112,263	413,657
Positive Score Rate	64.01% (71,869 images)	76.09% (314,771 images)

Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 2 is:

Participation History	Age Factor	Bonus
Phase 3 — Cycle 1 (and active now)	1.30	+30%
Phase 4 — Cycle 1 (and active now)	1.20	+20%
Phase 4 — Cycle 2 (and active now)	1.10	+10%
Phase 4 — Cycle 3 (new miner)	1.00	+0%

Final Score Calculation

We're using the same formula as last cycle:

Total Score = (Remaining Rewards from Phase 4 – Cycle 1) + (Face Variation Score × Age Factor)

Where:

- Remaining Rewards from Previous Cycle carries forward any unrealized rep_score from Phase 4 — Cycle 1.
- Face Variation Score is computed from the Δrep mapping table (unchanged, see below).
- Age Factor is `1.20`, `1.10`, or `1.00` based on the participation history described above.

Tier (Extended Top-End Normalization)

Tiers stay the same as last cycle, including the Platinum tier introduced in Phase 4 — Cycle 2. For reference, the table is included again below:

Tier	Range of Score	Multiplier
Platinum	200+	1.25
Diamond	50–200	1.15
Gold	10–49.99	1.10
Silver	2.00–9.999	1.05
Bronze	1.00–1.999	1.02

☐ Neutral	0.70–1.00	1.00
‼ Watch	0.10–0.699	0.90

Decay

Decay will increase from 0.02 -> 0.05

Leaderboard Growth

As reported in the Cycle 3 mid-cycle update, this cycle saw a major jump in output quality and volume even as the number of unique participants held roughly steady. Total submissions rose from 112,263 to 413,657, the positive score rate climbed from 64.01% to 76.09%. A quick recap of those headline numbers:

Metric	Phase 4 – Cycle 2	Phase 4 – Cycle 3
Total Submissions	112,263	413,657
Positive Score Rate	64.01%	76.09%

Now that final scoring (including the similarity penalty) is complete, the reputation leaderboard for Phase 4 — Cycle 3 holds 89 miners. This shows an overall average increase in miner quality. More miners are getting higher tiers. The tier breakdown for this cycle:

Tier	Cycle 2	Cycle 3
Platinum	1	1
Diamond	4	20
Gold	3	10
Silver	1	13
Bronze	7	11
Watch	7	7

Because of the size of the leaderboard, only the top 10 miners are listed below.

Miner's Reputation Leaderboard (Top 10)

Reputation is determined by validated Face performance. Miners not listed here fall into the Neutral tail or the Watchlist, or did not score high enough to surface.

Rank	Hotkey	Score	Tier
1	5Hmz...BE63	309.75	Platinum
2	5DUy...U39a	146.82	Diamond
3	5GsC...Kybn	128.35	Diamond
4	5Cd4...3sMU	125.14	Diamond
5	5DCy...V5N6	102.26	Diamond
6	5C7X...aSVT	90.75	Diamond
7	5FqF...8Pro	86.86	Diamond
8	5H8o...6xVv	85.97	Diamond
9	5FCD...Z6fa	74.03	Diamond
10	5HU4...4dyu	73.56	Diamond

Δrep Mapping Table (unchanged)

Validation Score	Meaning	Δrep
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03
+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
–1	Partial spam	-0.05
–2	Misleading	-0.08
–3	Duplicate	–0.10
–4	Not enough info	-0.25
–5	Cheat	-0.5

Top 3 Miners' Score Breakdown

The tables below show how raw validation outcomes translated into reputation for the top three miners this cycle.

5Hmz...BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	1	0	3	55	188	444	839	1785	182	3497
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	-0.1	0	-0.15	0	1.88	8.88	25.17	71.4	18.2	125.28

Face score = 125.28

Last cycle score = 146.89

Age = 4 = 1.3 (age factor)

Age score = 125.28 × 1.3 = 162.86

Rep score = last cycle score + age score = 309.75

5DUy...U39a

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	1	1	11	50	310	654	945	960	8	2940
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	-0.1	-0.08	-0.55	0	3.10	13.08	28.35	38.4	0.8	83

Face score = 83

Last cycle score = 38.92

Age = 4 = 1.3 (age factor)

Age score = 83 × 1.3 = 107.9

Rep score = last cycle score + age score = 146.82

Phase 4, Cycle c3(update) – Jun 5, 2026

Post-Phase 4 — Cycle 3 Updates (June 5, 2026)

Cycle 3 Summary: Miner Participation and Data Quality

One of the primary goals heading into Phase 4 — Cycle 3 was to increase miner participation and to raise the overall quality of submitted data. Both objectives were achieved. The cycle saw meaningful growth in the volume of high-quality submissions, with miners producing more accurate and diverse face variations than in previous cycles. These results reflect improvements across the network and demonstrate that the ecosystem is maturing in a positive direction.

Below is a high-level snapshot of the key participation and performance metrics for this cycle.

Metric	Phase 4 – Cycle 2	Phase 4 – Cycle 3
Cold Keys (Unique Miners)	26	24
Hot Keys (Active)	101	74
Total Predicted Score	889.27	3,798.81
Total Challenges	4,930	3,455
Total Submissions	112,263	413,657
Positive Score Rate	64.01% (71,869 images)	76.09% (314,771 images)

Predicted Score Improvement — A Strong Signal, With a Caveat

The predicted score for Phase 4 — Cycle 3 stands at 3,798.81, compared to 889.27 in the previous cycle — a jump of +327.2%. This is a very encouraging result and reflects the significant effort miners have put into improving the quality and consistency of their submissions. The top performer this cycle, 5FCDcRLjtnHx...Z6fa, achieved a predicted score of 143.36, an improvement of over 1,087% from the prior cycle.

It is important to note, however, that these are predicted scores and do not yet account for post-processing penalties. In particular, similarity-based penalties — which cross-reference submissions against the full image database to detect near-duplicate or copy-paste outputs — have not yet been applied. Once these penalties are factored in, some individual scores may decrease. The final scores will be published after the full post-processing pipeline is complete.

Miners should therefore treat the numbers in this update as a strong directional indicator rather than a final settled result.

Miner Score Improvement (Returning Miners)

Hotkey	Predicted Score (C3)	Score (C2)	% Change
5HmzMBNb…BE63	165.59	193.01	-14.2%
5FCDcRLj…Z6fa	143.36	12.07	+1,087.7%
5C5hoeGp…kAQF	115.18	74.55	+54.5%
5EBxQcMK…LAqt	112.58	49.39	+127.9%
5DUyTmjJ…U39a	110.09	136.42	-18.9%
5GsChcZ5…Kybn	109.69	128.08	-14.4%
5FswNJmT…juX9	107.34	1.00	+10,612%
5FqFYjSk…8Pro	92.08	108.61	-15.2%

Challenge Bitmind's AI Detection — Early Progress

One of the objectives this cycle was for miners to begin producing face variations that can bypass Bitmind's AI face detector. While the network has not yet cracked it at scale, there are early and encouraging signs of progress.

Across all face submissions, 5,787 images (1.93%) were flagged as non-AI by Bitmind's face detector — meaning those images successfully passed as real. The top-performing miner in this category, 5DUyTmjJ…U39a, achieved a 13.81% is-AI-false rate across 8,088 images. Several other miners also showed meaningful rates in the 10–14% range.

Hotkey	Total Sent	Is-AI-False	Is-AI-False %
5DUyTmjJ…U39a	8,088	1,117	13.81%
5CCeGaBJ…Mnr	6,294	864	13.73%
5Gy5RRZ6…Lci	5,295	716	13.52%
5GeXs96K…CGL	4,647	598	12.87%
5HY9pKAQ…Pd8	3,025	323	10.68%

On the screen detector side, 722 images (2.56%) passed as non-AI out of 28,241 evaluated. The top performer was 5Hp6G4j7…vQos with a 15.28% false rate across 72 submissions. Only 33% of each miner's images were randomly sampled for this evaluation, so results represent a sample of the full pool.

This is a very encouraging start. Consistently breaking AI detection at scale remains one of the key goals for the next cycle, and miners who develop strong techniques in this area will be well positioned going forward.

Copy-and-Paste Exploit — This Must Stop

During post-processing for this cycle, the team identified cases where miners submitted face variations that were produced by simply cutting out the subject from the original seed image and pasting it onto a different background. This is a shortcut that does not constitute a genuine transformation and will not be tolerated.



Miners found to have participated in this exploit will receive a lower score as a result. As noted in Section 2, these penalties have not yet been applied to the predicted scores above — but they will be. If you have been submitting copy-and-paste variations, be aware that this will negatively affect your final score for this cycle.

The network rewards originality and genuine transformations. Shortcuts undermine the value of the dataset and the integrity of the validation pipeline.

Summary and Thank You

Phase 4 — Cycle 3 has been one of the strongest cycles to date. Miner participation improved, predicted scores jumped by over 327%, and the first meaningful signs of AI detection bypass have emerged. These are the results the team has been working toward, and they reflect real effort from the mining community.

To summarize the key takeaways from this cycle:

- Participation goals were met: more miners, more high-quality submissions.
- The total predicted score rose by +327.2% — a strong signal of improved data quality across the network.
- These are predicted scores only. Post-penalties (including similarity and copy-paste checks) have not yet been applied and may adjust individual results downward.
- AI detection bypass is showing early promise. The next cycle will place an even greater emphasis on this goal.
- Copy-and-paste submissions have been identified and will be penalized. This behavior must stop.

Thank you to every miner who participated in Phase 4 — Cycle 3. Your contributions are what make this network valuable. Keep mining, keep improving, and we look forward to sharing final results and cycle 4 details soon.

Post-Phase 4 — Cycle 2 Updates (April 20, 2026)

Phase 4, Cycle c2(end) – May 2, 2026

Status of Manual Grading

As of April 20, 2026, the Face Variations collection phase for Phase 4 — Cycle 2 has officially concluded. Since that date, the team has shifted full focus to the manual grading pipeline for Faces and to the integration of screen replays into the validation workflow.

At this point, all submitted Face Variations have been processed through the grading workflow. The same evaluation criteria from the previous cycle continue to apply — the steps remain unchanged for consistency and fairness:

Step 1: Filter out all corrupt and duplicate images.

Step 2: Confirm identity is relatively preserved (AdaFace).

Step 3: Run the image through variation-type detection.

Step 4: Manual validator checks for:

- Is the person still the same individual?
- Did the miner follow the query?
- Did the miner exploit?
- Does the image have more than one variation?
- Are artifacts, overlays, or shortcuts present?

Screen replays were added as a new validation signal in this cycle. Replays let the team verify miner workflows end-to-end during manual review, and the results have been very positive — they made it noticeably easier to confirm that variations were genuinely produced rather than copy-pasted, looped, or otherwise shortcut.

Note: This cycle does not include UAVs. Final scoring is therefore based entirely on the Face Variation evaluation.

Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 2 is:

Participation history	Age Factor	Bonus
Phase 3 — Cycle 1 ****and**** Phase 4 — Cycle 1 (and active now)	1.20	+20%
Phase 4 — Cycle 1 only (and active now)	1.10	+10%

The bonus is incorporated through the Age Factor in the score formula below.

Final Score Calculation

Because there is no UAV component this cycle, the formula simplifies to:

Total Score = (Remaining Rewards from Phase 4 – Cycle 1) + (Face Variation Score × Age Factor)

Where:

- Remaining Rewards from Previous Cycle carries forward any unrealized rep_score from Phase 4 — Cycle 1.
- Face Variation Score is computed from the Δrep mapping table (unchanged, see below).
- Age Factor is `1.20`, `1.10`, or `1.00` based on the participation history described above.

Slight Change in the Rewards Mechanism

We made two adjustments to the reward mechanism for this cycle: a new Platinum tier and a faster decay rate. Full technical detail lives in the UAV rewards report; the summary is below.

New Platinum tier (extended top-end normalization)

Tier	Multiplier
🥇 **Platinum**	**1.25**
💎 Diamond	1.15
⭐ Gold	1.10
🪙 Silver	1.05
🥉 Bronze	1.02
⬜ Neutral	1.00
‼️ Watch	0.90

The normalization curve was also extended at the top so that high-rep miners are no longer compressed near the cap. Platinum miners now have explicit headroom (raw 200+ → normalized 2.00–3.00).

Faster decay

Decay per query: 0.01 -> 0.02

Why we changed this

Over the last two cycles, some miners did not receive their full Face variation rewards within the cycle window. We carried the remainder forward by adding it to their score in the following cycle. In practice we found this unfair: rolling unpaid rewards into future scores meant some miners might never fully clear what they were owed in a single cycle, and the accounting felt perpetual rather than settled.

Increasing rep-score decay shortens the reward horizon and aligns it with how often miners are evaluated. That is fairer to miners who actually earned rewards in the current window — emissions are realized on a clearer timetable instead of

being indefinitely deferred through score carryover. It also prevents the leaderboard from freezing in place: top miners cannot stay dominant forever purely from slow score drainage, and after a cycle ends, everyone starts from a more level playing field for the next one.

Leaderboard Growth

Phase 4 - Cycle 2 saw a substantial increase in miner participation. The leaderboard is significantly larger and more diverse than the previous cycle:

Metric	Phase 4 - Cycle 1	Phase 4 - Cycle 2
Miners on the leaderboard	8	**100**

Tier breakdown for Phase 4 - Cycle 2:

Tier	**Count**
Platinum	1
Diamond	4
Gold	2
Silver	1
Bronze	7
Neutral	78
Watch	7

Because of the size of the leaderboard, only the **top 15** miners are listed below

Miner's Reputation Leaderboard (Top 15)

Reputation is determined by validated Face performance. Miners not listed here either fall into the long Neutral tail, the Watchlist, or did not score high enough to surface.

Rank	Hotkey	Score	Tier
1	5Hmz	285.95	Platinum
2	5DUy	161.89	Diamond
3	5GsC	140.88	Diamond
4	5FqF	108.61	Diamond
5	5C5h	63.71	Diamond

6	5Cev	45.67	Gold
7	5EBx	35.49	Gold
8	5GP1	28.88	Silver
9	5FCD	11.62	Bronze
10	5CS3	11.18	Bronze
11	5Dks	8.50	Bronze
12	5Fzs	8.43	Bronze
13	5GQr	8.19	Bronze
14	5GZd	6.30	Bronze
15	5EPi	5.02	Bronze

Δrep Mapping Table (unchanged)

Validation score	Meaning	Δrep
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03
+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
–1	partial spam	-0.05
–2	Misleading	-0.08
–3	Duplicate	–0.10
–4	Not enough info	-0.25
–5	Cheat	-0.5

Top 2 Miner's Score Break

The following table shows how raw UAV outcomes translated into reputation for the top two miners. Each category multiplies by weight depending on the severity or quality of the submission.

Hotkey: 5Hmz…BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	1	3	2	13	76	355	1177	2509	498	4634
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	-0.1	-0.24	-0.1	0	0.76	7.1	35.31	100.36	49.8	192.89

Face score = 192.89

Last cycle score = 54.48

Age = 3 = 1.2 (age factor)

Age score = 192.89 * 1.2 = 231.47

Rep score = last cycle score + age score = 285.95

Hotkey: 5GsC… Kybn

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	0	3	9	200	484	739	1024	1614	207	4280
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	0	-0.24	-0.45	0	4.84	14.78	30.72	64.56	20.7	134.91

Face score = 134.91

Last cycle score = 0

Age = 3 = 1.2 (age factor)

Age score = 134.91 * 1.2 = 161.89

Rep score = last cycle score + age score = 161.89

Score Normalization Ranges

Each tier has a bounded window mapping raw reputation into a reward-safe range.

Tier	Raw Range	Normalized Range
Watch	0.10–0.699	0.50–0.70
Neutral	0.70–1.00	0.70–1.00
Bronze	1.00–1.999	1.00–1.20
Silver	2.00–9.999	1.20–1.50
Gold	10–49.99	1.50–1.80
Diamond	50–200	1.80–2.00
Platinum	200-999	2.00-3.00

Weight Shift: KAV vs Face Variation/UAV

We also updated the score weighting to put more emphasis on adversarial image performance:

- KAV weight: 0.15 → 0.10
- Face Variation/UAV weight: 0.85 → 0.90

Previous formula:

Miner_rep = 0.85 · Δface_score + 0.15 · Δrep_KAV

New formula:

$$Miner_rep = 0.90 \cdot \Delta face_score + 0.10 \cdot \Delta rep_KAV$$

This change reflects Phase 4 priorities: stronger focus on Face Variation and UAV signal, and less emphasis on baseline KAV contribution.

Score Sheet

Score	Condition / path	Status
−5	Duplicate-level (copy and paste images into different backgrounds instead of synthetic generation)	rejected
−4	Severe aspect-ratio distortion (variation face >15.5% narrower than seed; delta < −0.155)	rejected
−3	Face identity below threshold or duplicate with other images (based on hash); or MediaPipe face count is not exactly 1 (0 or >1); religious head covering + gender-specific request but seed gender mismatch	recommended_scored / rejected
−2	No variation detected (claimed variation type but comparison detected no variation); or duplicate with other images (duplicate similarity checker)	rejected
−1	Duplicate with seed image (hash match); or variation resolution ≤ 512x672; or face not dominant part of image; or background accessory path with insufficient match and no background detected (0−1 = −1)	recommended_scored / rejected
0	Download/write failed; comparison failed (manual review); label mismatch with variation detected (rejected); background accessory path insufficient match	rejected / manual_review / recommended_scored
2	Background accessory path: would be 3 but no background change detected (3−1)	recommended_scored
3	Standard: identity 0.6–0.95 with partial variation match; background accessory: conf ≥ 0.5 (matching or other accessory)	recommended_scored
4	Background accessory: would be 5 but no background detected (5−1)	recommended_scored
5	Standard: identity ≥ 0.6 with full variation match; background accessory: matching accessory + ada_sim ≥ 0.6 + conf ≥ 0.5	recommended_scored

Screen replay score sheet:

Score	Condition / path	Status
Info	**The 3 checks are:** cue 1 present, cue 2 present, and device present.	reference
−3	*screen_replay*: face identity below threshold or duplicate with other images (based on hash); or MediaPipe face count is not exactly 1 (0 or >1);	recommended_scored / rejected
−2	*screen_replay*: duplicate with other images (duplicate similarity checker)	rejected
−1	*screen_replay*: duplicate with seed image (hash match); or variation resolution ≤ 512x672	recommended_scored / rejected
0	*screen_replay*: download/write failed; comparison failed (manual review); label mismatch with variation detected (rejected); identity < 0.4 (hard gate); all 3 checks missing; or detection failed (manual review)	rejected / manual_review / recommended_scored
1	*screen_replay*: identity ≥ 0.6 with only 1 of the 3 checks present (2 missing)	recommended_scored
3	*screen_replay*: identity ≥ 0.6 with 2 of the 3 checks present (1 missing)	recommended_scored
5	*screen_replay*: identity ≥ 0.6 and all 3 checks pass (cue 1 + cue 2 + device)	recommended_scored

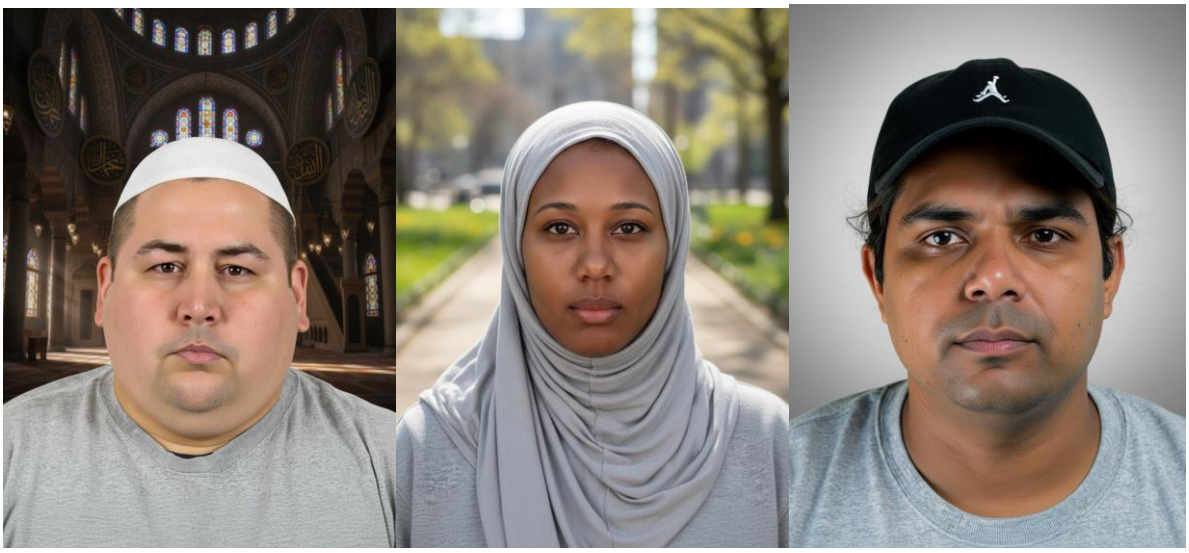
Examples

Screen reply





Background



Pose



Lighting



Expression



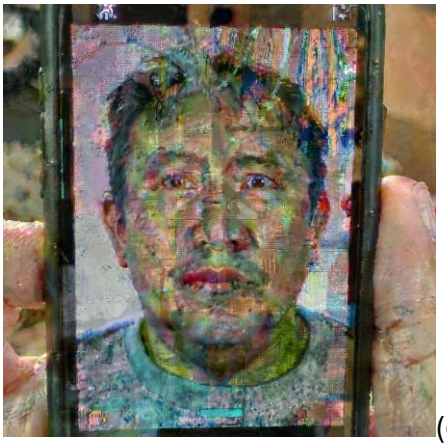
Negative examples (common pit-falls)


(Face is not dominant)


(No accessories)

   (Have more then one accessory and gender does not align)

 (Gender miss match (women do not typical wear turbans))

 (color disfiguration to the point where face not detect)

Phase 4, Cycle c1(end) – Mar 14, 2026

Post-Phase4-Cycle 1 – Updates (March 2, 2026)

Status of Manual Grading

As of March 2, 2026, the Face Variations collection phase has officially concluded. Since that date, the team has shifted its full focus to completing the manual grading pipeline for all submitted Face Variations.

At this point, all submitted Face Variations have been processed through the grading workflow. The same evaluation criteria previously outlined in earlier updates—referenced in other sections of this document—continue to apply throughout the grading process. These criteria ensure consistency, fairness, and alignment with the project's overall validation standards.

In earlier updates, we provided an overview of how the manual validation pipeline operates. This section expands on that explanation by providing additional details on scoring progress and by introducing the reputation tiers that will be used to determine emissions.

Within this document, individual Face scores will be released, along with the combined total score calculated from both the Face evaluation and the associated UAV address score. This combined score determines the final reputation adjustment.

For reference, the calculation formula used to determine the final reputation change is as follows:

$$\Delta rep_{p4} = 0.95 \cdot \Delta rep_{face} + 0.05 \cdot \Delta rep_{UAV_address}$$

This formula reflects the weighting of the two components:

- 95% of the final reputation score is derived from the Face evaluation.
- 5% of the final reputation score is derived from the UAV address reputation.

This weighted approach ensures that the quality and validity of the submitted Face Variations remain the primary factor in scoring, while still incorporating a smaller contribution from the associated address reputation.

Time-Based Rewards and Reputation

In this cycle, we are introducing a new feature designed to reward miners who have consistently contributed high-quality, non-exploitative data over an extended period of time. This feature recognizes long-term participation and sustained contributions to the network.

Any miner who participated in Phase 3 – Cycle 1 and Phase 4 – Cycle 1 will receive a 10% bonus applied to their score. This adjustment reflects the value of continued participation and reliable data contributions across multiple phases of the project.

The bonus is incorporated into the overall scoring system through an age factor, which increases the final score for qualifying miners.

Final Score Calculation

The total score used to determine rewards and reputation adjustments is calculated using the following formula:

Total Score = (Remaining Rewards from Phase 3 – Cycle 1) + ((UAV Address Rewards × 0.05) + (Face Variation Score × 0.95)) × Age Factor

This formula maintains the same weighting structure described earlier in the document:

- 95% of the score is derived from the Face Variation evaluation.

- 5% of the score is derived from the UAV address reputation.

The Age Factor is applied afterward to reward miners who participated in both Phase 3 – Cycle 1 and Phase 4 – Cycle 1, providing a 10% bonus to their final score.

Miner's Reputation Leaderboard

Reputation is determined by validated Face performance. Miners not listed here fall into the Normal, barely participated, or Watchlist, with negative or near-zero scores due to spam, duplicates, invalid coordinates, or consistently low-value UAVs.

Rank	hotkey	score	tier
1	5HmzMBNbEgjTxzxzbpg42hkrM5gzvPcaApD1kKP5nEq8BE63	115.97	💎 Diamond
2	5GsChcZ5FBfdbJRX4cTRgke5upQ3tQ24Z1K3r7PgojD9Kybn	28.87	🥈 Silver
3	5DUyTmjJ5P2xFHg6qBJW4FkbjKrwLwg7SvU8UcwdqYiEU39a	8.36	🥉 Bronze
4	5CS3nxrpwCkrbv3PYQD6LLDtxEickHq21sJ9FPAvewkYpouY	5.59	🥉 Bronze
5	5GZdCsRkiXJ5Kcks7iXZhAgC4aLna3r5JUhcJkHkL1RTPr51	2.94	▯ ◯ Neutral
6	5C7MdoMGTpgRBNEodu6XkNWJn7zJHAmAKnz4VjLJUL8JmT3D	2.35	◯ Neutral

7	5G4jmgfWjapk4amvVCciebCqav4yPjtR3Jp3ZJUJ3d6beeKG	2.38	○ Neutral
8	5DFmJzd32EqKJFT9XUZFYDB6MPahQDP38WHACeToNGQ2NK5e	2.05	○ Neutral

Top 2 Miner's Score Break

The following table shows how raw UAV outcomes translated into reputation for the top two miners. Each category multiplies by weight depending on the severity or quality of the submission.

Hotkey: 5Hmz...BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	18	0	3	175	37	499	634	577	1572	377	600	4442
Δrep	0.5	0.25	0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	-9	0	-0.3	-14	-1.85	0	6.34	11.5	47.2	15	60	115.97

remaining rewards from phase 3 cycle 1 = 0

UAV address rewards = -171.2

Face variation = 115.97

age factor = 1.1

Total score: **111.77** = (0) + ((-171.2 *0.05) + (115.97 * 0.95) * 1.1)

Hotkey: 5GsC... Kybn

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	4	9	18	121	217	159	304	193	77	1102
Δrep	0.5	0.25	0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	

Total	0	0	-0.4	-0.72	-0.9	0	2.17	3.18	9.12	7.72	7.7	**28.87**

remaining rewards from phase 3 cycle 1 = 0

UAV address rewards = 0

Face variation = 28.87

age factor = 1

Total score: 27.43 = (0) + ((0 *0.05) + (28.87 * 0.95) * 1)

The table below, Δrep Mapping Table, shows how each validation score contributes to a miner's reputation, with positive scores increasing Δrep and negative scores reducing it.

Validation score	Meaning	Δrep
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03
+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
–1	partial spam	-0.05
–2	Misleading	-0.08

–3	Duplicate	–0.10
–4	Not enough info	-0.25
–5	Cheat	-0.5

Reputation Tiers & Normalization (How Miner Rewards Are Calculated)

Reputation is not used raw. Instead, miners are assigned a tier and their scores are normalized before being applied to emissions. This ensures fairness and prevents extremely high scores from dominating the incentive pool.

Tier Multipliers

These increase or decrease a miner's effective reputation:

Tier	Multiplier
💎 Diamond	1.15
⭐ Gold	1.10
🪙 Silver	1.05
🥉 Bronze	1.02
⬜ Neutral	1.00
‼️ Watch	0.90

Score Normalization Ranges

Each tier has a bounded window mapping raw reputation into a reward-safe range.

Tier	Raw Range	Normalized Range
Watch	0.10–0.699	0.50–0.70
Neutral	0.70–1.00	0.70–1.00

Bronze	1.00–1.999	1.00–1.20
Silver	2.00–9.999	1.20–1.50
Gold	10–49.99	1.50–1.80
Diamond	50–9999	1.80–2.00

The final Face/UAV reward is Normalized Score × Tier Multiplier, which is then will be the 80% of the final score, combined with KAV quality (20%), and the final score will be the score the validator will set weight on the chain based on.

Below is a **clean section written to match the tone and structure of the rest of your documentation**. It keeps the same explanatory style, avoids casual language, and clearly explains both **what was discovered** and **how duplicates were handled**.

Duplicate Submissions and Multi-Hotkey Activity

During the manual validation and post-processing stage of the Face Variations pipeline, the team also performed cross-miner duplication analysis. This process was designed to detect situations where the same underlying model or workflow was producing nearly identical outputs across multiple miners.

The goal of this analysis is to ensure that the network rewards **unique and independently generated data**, rather than scaled replication of the same output across multiple miners.

Real Example From Cycle 1

A clear example illustrates why this policy is necessary.

In Cycle 1, a single operator ran five separate hotkeys (5GsC, 5EWg, 5HZF, 5FyA, and 5ERu). All five performed well in the initial scoring results and each reached the Silver tier, with scores of 28.87, 28.57, 28.27, 26.10, and 25.53, placing 2nd through 6th on the leaderboard.

These scores placed the five hotkeys **2nd through 6th on the leaderboard**.

If these submissions had been considered independent and unique, the five hotkeys combined would have earned approximately **17,942 α over roughly 40 days** as their scores gradually decayed through validator reward allocation.

Duplication Detection Results

However, after running the duplication detection pipeline, we discovered that the five hotkeys were submitting **nearly identical images**.

In practice, this meant the same model was producing the same outputs and distributing them across multiple miners. Because the images were not uniquely generated for each submission, the system classified these results as **duplicate data**.

Duplicate data undermines the purpose of the validation pipeline, which is designed to reward **unique transformations and independent contributions**.

Penalty Outcome

As a result, all five hotkeys were penalized and placed on the **Watch tier**, causing their reputation scores to drop to near zero.

In the subsequent leaderboard snapshot:

- Only one hotkey briefly remained visible with a score of **0.15**
- That hotkey earned approximately **34 α for a single day**
- After that, all five hotkeys effectively disappeared from the leaderboard

Why Running Multiple Hotkeys Backfired

Ironically, the miner would have earned significantly more by operating a **single hotkey**.

With their highest original score of **28.87**, a single miner would have:

- earned **higher daily rewards**, since they would not be competing with their own duplicated outputs
- remained eligible for **40+ days** as reputation gradually decayed
- likely earned **approximately 8,500–10,000 α** in total rewards

Instead, by splitting the same model across five hotkeys and triggering duplication detection, the miner's expected outcome shifted from a realistic **~9,000 α** to **essentially 0 α**.

How Duplicate Submissions Were Handled

To handle these cases fairly, the system applied a deterministic duplicate resolution policy.

When identical or near-identical submissions were detected across multiple miners:

- **One hotkey was selected to retain the positive score that comes under the same cold key as the other hotkeys that submitted duplicate data.**
- The **duplicate submissions were assigned negative scores**

Importantly, the system **consistently selected the same hotkey** to keep the score rather than randomizing the selection. This approach was intentionally lenient.

Without this rule, duplicate submissions could have resulted in **all participating hotkeys receiving negative outcomes**, leaving the miner with no rewards at all.

Instead, the system allowed **one hotkey to retain valid rewards**, while penalizing the duplicates.

Policy Going Forward

This leniency should not be assumed in future cycles.

The current approach was chosen to avoid fully eliminating rewards for miners who had already contributed valid data under at least one hotkey. However, future duplication handling may adopt **randomized or stricter enforcement**, which could result in **all duplicated hotkeys receiving negative scores or no rewards**.

For this reason, miners are **strongly encouraged to operate a single hotkey when generating Face Variations**.

Submitting the same model outputs across multiple hotkeys will be treated as duplication and may lead to **watchlisting, reputation penalties, and the loss of emissions**.

Summary

One miner producing **high-quality, unique data** will earn more rewards over time.

Multiple miners running the **same model and submitting duplicated outputs** will trigger duplication detection, leading to **penalties, watchlisting, and loss of rewards**.

The incentive structure is therefore designed to reward **originality and sustained high-quality contributions**, not horizontal scaling of identical outputs.

Participation Policy (Sandbox vs Execution)

Phase 4 — Cycle 2 Sandbox (current stage)

- During the Sandbox stage, Face/UAV reputation is not used for emissions yet. Validators continue operating normally, but Face/UAV-based reward allocation is not activated in this stage.
- Sandbox is for calibration and stability checks before Face/UAV rewards go live.

Phase 4 — Cycle 2 Execution (UAV rewards go live)

- Once Execution begins, validators will start allocating the Face/UAV portion of rewards using the Phase 4 — Cycle 1 reputation results (each miner's rep_score + tier).
- At that point, miners with higher reputation tiers (e.g., Diamond) will receive stronger and longer-lasting UAV rewards because their rep_score starts higher and their tier multiplier is higher.

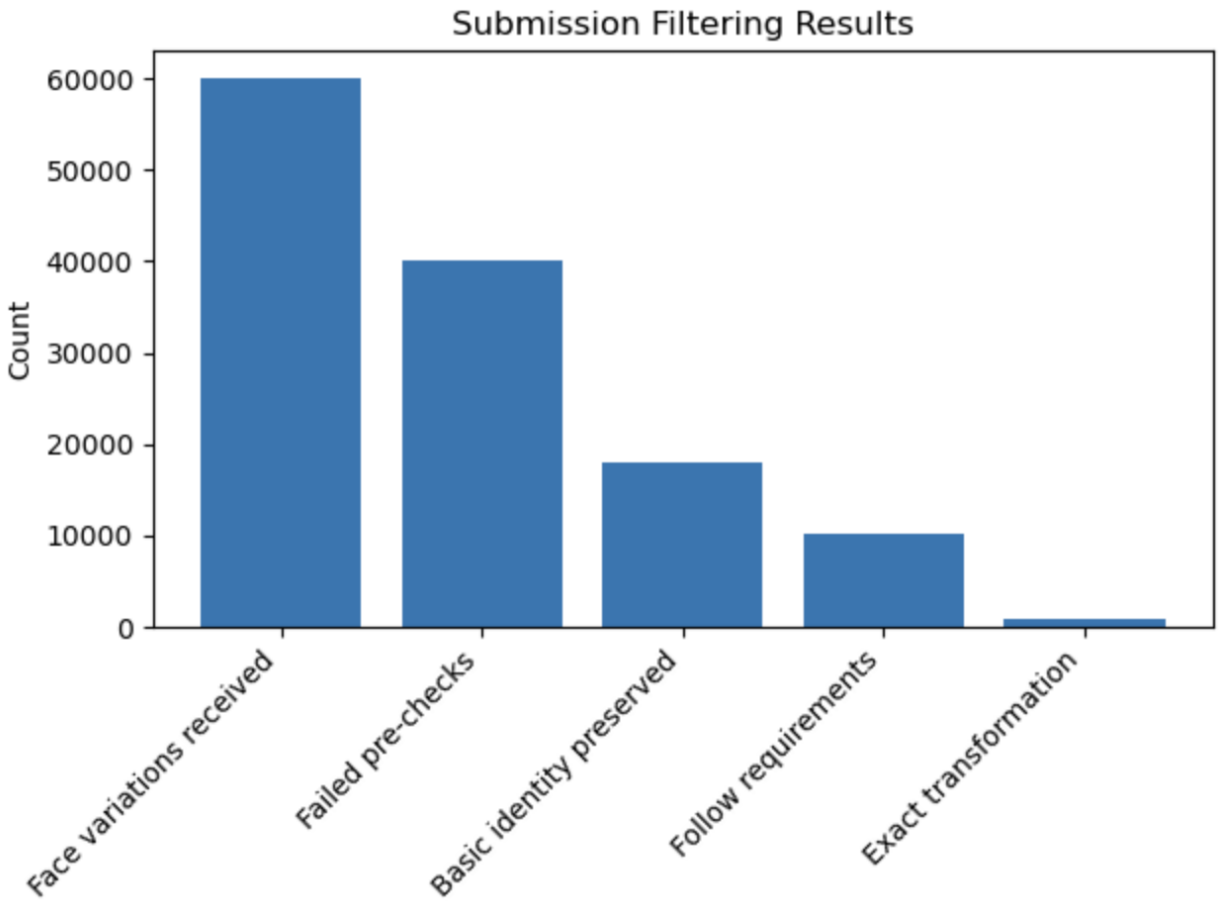
Reputation decay (applies once UAV rewards are live)

- Every time a miner receives a validator reward allocation, their rep_score will be reduced by 0.01 in the database.
- At the end of Phase 4 — Cycle 2 Execution, any remaining rep_score (if > 0) will be carried forward and combined with the next cycle's reputation update. Face/UAV-based rewards will resume in Phase 4 — Cycle 3 Execution, using the updated reputation snapshot.
- This means Watch-tier miners will decay to 0 faster and stop receiving Face/UAV rewards sooner, while higher-tier miners will remain eligible longer—unless they keep contributing validated Face/UAVs to rebuild score.

Tier & score rules

- A miner's **tier is fixed within a cycle** and only updates when the next UAV cycle snapshot is computed.
- rep_score is dynamic and will **trend downward through decay** as rewards are paid out.

Statistics on the Face variations



Face variations are currently one of the highest-load pipelines in the system. Their purpose is to generate constrained biometric transformations such as pose, lighting, expression, or background edits while preserving the identity of the validator-provided subject.

Like UAVs, this workflow remains heavily manual, and the volume illustrates why strong filtering and reputation mechanisms are essential.

Over **60,000 Face variations** were received.

More than **40,000** entries failed automatic pre-checks due to:

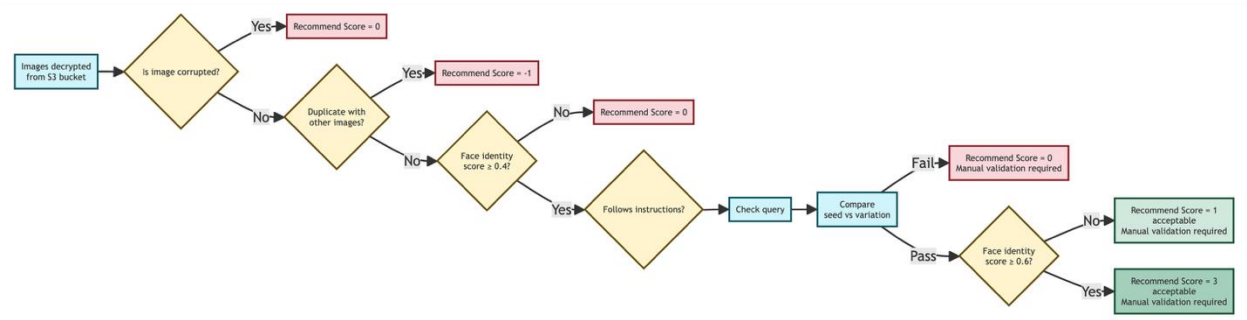
- corrupted or unreadable images

- duplicate of the original seed image
- duplicate of another miner's submission

More than **18076** preserve identity at a basic recognizable level
Nearly **10267** follow the query requirements *and* preserve identity
More than **947** perform exactly the requested transformation, no more and no less, while preserving the identity of the seed image.

Flow chart on how Face Validation are Processed

Below is a small window into how manual validation is performed for face variation submissions.



Step 1: Filter out all the corrupt and duplicate images

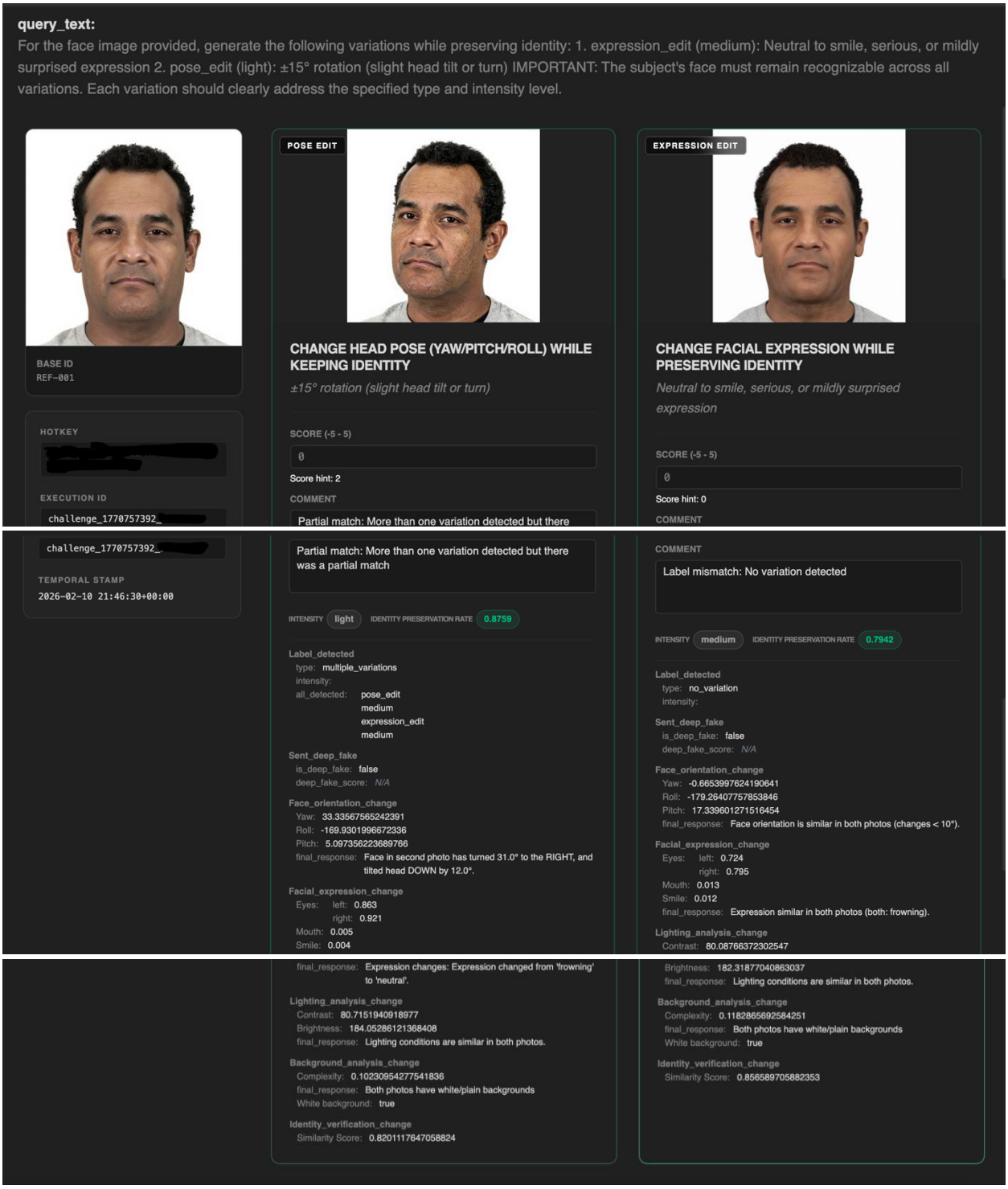
Step 2: Identity is relatively preserved (AdaFace)

Step 3: Run the image to detect the variation type

Step 4: Sent to the manual validator checks for:

- Is the person still the same individual?
- Did the miner follow the query?
- Did the miner exploit?
- Does the image have more than variations?

- Are artifacts, overlays, or shortcuts present?



(Here is a sneak peak on what the manual validator sees)

Examples Of Face Variations Processed:

Rejected Identity not Perserved:

1. Seed Image
2. Expression Edit (Face identity below threshold (ada_sim=0.322 < 0.4))
3. Background Edit (Face identity below threshold (ada_sim=0.313 < 0.4))
4. Lighting Edit (Face identity below threshold (ada_sim=0.308 < 0.4))



1. Base Image
2. Expression Edit (Face identity below threshold (ada_sim=0.33 < 0.4))



Duplicates that will be rejected starting from Feb 16th :

If a miner just **copies and pastes** the **face** on **a different background,** they will start to get negative scores.

Examples:

1. Seed Image
2. Background edit (copy and paste of the face on different background)



1. Seed Image
2. Background edit (copy and paste of the face on different background)



Positive cases that still need Manual Review:

1. Seed Image
2. Pose edit (Partial match: More than one variation detected but there was a partial match, detected: pose edit – far and expression edit – medium, score hint = 2)

3. Expression edit (Partial match: More than one variation detected but there was a partial match, detected: pose edit - medium and expression edit – far, score hint = 2)
3. Lighting edit (Acceptable (ada_sim=0.822, mediapipe_sim=0.626); Label close: claimed 'lighting_edit' 'far', detected 'medium', score hint = 3)

